

Problem-Solution Fit Canvas

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

Field	Details
1. CUSTOMER SEGMENT(S) [CS]	College students preparing for semester exams. School students preparing for board exams. Competitive exam aspirants. Self-learners studying independently.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	Difficulty understanding lengthy study materials (Daily). Manual note-making and flashcard creation is time-consuming (Daily). No self-assessment tools from own material (Weekly). Lack of personalized study plans (Every exam cycle). Scattered study resources across multiple apps (Daily).
3. TRIGGERS TO ACT [TR]	Approaching exam deadlines. Failing to retain information from long reading sessions. Seeing classmates use AI tools for study preparation. Frustration with manual flashcard creation.
4. EMOTIONS BEFORE / AFTER [EM]	BEFORE: Overwhelmed, anxious, frustrated, directionless, stressed. AFTER: Confident, relieved, organized, empowered, prepared.
5. AVAILABLE SOLUTIONS PROS & CONS [AS]	ChatGPT/Gemini directly: Pro — Free AI access. Con — No file upload, no persistence, no authentication, no structured output. Quizlet: Pro — Flashcard creation. Con — Manual input required, no AI generation from own material. YouTube summaries: Pro — Visual explanations. Con — Not personalized to student's own material.
6. CUSTOMER LIMITATIONS [CL]	Limited budget (students prefer free tools). Need internet access for AI API calls. Limited time during exam preparation periods. Basic tech proficiency (need simple API interface).
7. BEHAVIOR + ITS INTENSITY [BE]	Reading long chapters and highlighting manually (High intensity, daily). Procrastinating on difficult subjects due to lack of structure (Medium intensity). Cramming in the last few days before exams (High intensity, every exam). Searching for shortcuts and online summaries (Medium intensity).
8. CHANNELS OF BEHAVIOR [CH]	ONLINE: Postman API client, REST API endpoints, MongoDB Atlas cloud database, Google Gemini API. OFFLINE: Study material files (.txt, .md, .pdf), handwritten notes converted to text files.
9. PROBLEM ROOT CAUSE [RC]	The volume of study material exceeds the student's available time for manual processing. No existing tool combines AI summarization + flashcards + quizzes + study planning from uploaded content in one authenticated platform. Students lack expertise to create quality self-assessment materials.
10. YOUR SOLUTION [SL]	AI StudyBuddy — A RESTful backend API built with Node.js, Express, and MongoDB that allows authenticated students to upload study materials and uses Google Gemini AI to automatically generate: (1) Concise summaries, (2) Q&A flashcards, (3) MCQ quizzes, (4) Personalized study plans. All generated content is persisted in MongoDB and accessible anytime via secure JWT-authenticated API endpoints.